



2025 HTOP FINAL PROJECT

STOCHASTIC CALCULUS AND APPLICATION IN OPTION PRICING

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HTOP Final Project

§1.1 Symmetric Random Walk

To create a Brownian motion, we began with a symmetric random walk.

Definition 1.1.1: symmetric random walk

To construct a symmetric random walk, we can repeatedly toss a fair coin. Denote the successive results of the tosses by $\omega = \omega_1\omega_2\omega_3 \dots$

$$X_j = \begin{cases} 1 & \text{if } \omega_j = H, \\ -1 & \text{if } \omega_j = T, \end{cases}$$
$$M_k = \sum_{j=1}^k X_j, \quad k = 1, 2, \dots$$

And The process M_k is a random walk.

§1.1.1 Increments of Symmetric Random Walk

A random walk has *independent increments*. This means that if we choose nonnegative integers

$$0 = k_0 < k_1 < \dots < k_m,$$

the random variables

$$M_{k_1} - M_{k_0}, (M_{k_2} - M_{k_1}), \dots, (M_{k_m} - M_{k_{m-1}})$$

are independent. Each of these random variables is called an increment of the random walk and with expected value 0 and variance $k_{i+1} - k_i$.

$$M_{k_{i+1}} - M_{k_i} = \sum_{j=k_i+1}^{k_{i+1}} X_j,$$
$$\text{Var}(M_{k_{i+1}} - M_{k_i}) = \sum_{j=k_i+1}^{k_{i+1}} \text{Var}(X_j) = \sum_{j=k_i+1}^{k_{i+1}} 1 = k_{i+1} - k_i.$$

§1.1.2 Martingale Property for Symmetric Random Walk

To see that symmetric Random walk is a martingale, choose nonnegative integers $k < \ell$ and compute:

$$\begin{aligned} \mathbb{E}[M_\ell | \mathcal{F}_k] &= \mathbb{E}[(M_\ell - M_k) + M_k | \mathcal{F}_k] \\ &= \mathbb{E}[M_\ell - M_k | \mathcal{F}_k] + \mathbb{E}[M_k | \mathcal{F}_k] \\ &= \mathbb{E}[M_\ell - M_k | \mathcal{F}_k] + M_k \\ &= \mathbb{E}[M_\ell - M_k] + M_k = M_k. \end{aligned}$$

§1.1.3 Quadratic Variation of Symmetric Random Walk

The quadratic variation up to time k is defined to be

$$[M, M]_k = \sum_{j=1}^k (M_j - M_{j-1})^2 = k.$$

Note that this is computed path-by-path. The quadratic variation up to time k along a path is computed by taking all the one-step increments $M_j - M_{j-1}$ along that path, squaring the increments and summing them.

§1.1.4 Scaled Symmetric Random Walk

To approximate a Brownian motion, we speed up time and scale down the step size of a symmetric random walk. More precisely, we fix a positive integer n and define the scaled symmetric random walk

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} M_{nt},$$

provided that nt is itself an integer. We shall obtain a Brownian motion in the limit as $n \rightarrow \infty$.

Theorem 1.1.1: Central Limit Theorem

Fix $t \geq 0$. As $n \rightarrow \infty$, the distribution of the scaled random walk $W^{(n)}(t)$ evaluated at time t converges to the normal distribution with mean zero and variance t .

Proof. For the normal density

$$f(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$$

with mean zero and variance t , the moment-generating function is

$$\varphi(u) = \int_{-\infty}^{\infty} e^{ux} f(x) dx = e^{\frac{1}{2}u^2 t}$$

The moment-generating function for $W^{(n)}(t)$ is

$$\varphi_n(u) = \mathbb{E} e^{uW^{(n)}(t)} = \mathbb{E} \exp \left\{ \frac{u}{\sqrt{n}} M_{nt} \right\} = \mathbb{E} \prod_{j=1}^{nt} \exp \left\{ \frac{u}{\sqrt{n}} X_j \right\}.$$

Because the random variables are independent, the right-hand side of may be written as

$$\prod_{j=1}^{nt} \mathbb{E} \exp \left\{ \frac{u}{\sqrt{n}} X_j \right\} = \prod_{j=1}^{nt} \left(\frac{1}{2} e^{\frac{u}{\sqrt{n}}} + \frac{1}{2} e^{-\frac{u}{\sqrt{n}}} \right) = \left(\frac{1}{2} e^{\frac{u}{\sqrt{n}}} + \frac{1}{2} e^{-\frac{u}{\sqrt{n}}} \right)^{nt}.$$

Consider the logarithm of $\varphi_n(u)$ and by change of variable $x = \frac{1}{\sqrt{n}}$ and applying the L'Hôpital's Rule, we can show that

$$\log \varphi_n(u) = nt \log \left(\frac{1}{2} e^{\frac{u}{\sqrt{n}}} + \frac{1}{2} e^{-\frac{u}{\sqrt{n}}} \right)$$

converges to $\log \varphi(u) = \frac{1}{2}u^2 t$, finishing the proof. ■

§1.2 Brownian Motion

Definition 1.2.1: Brownian Motion

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For each $\omega \in \Omega$, suppose there is a continuous function $W(t)$ of $t \geq 0$ that satisfies $W(0) = 0$ and that depends on ω . Then $W(t)$, $t \geq 0$, is a Brownian motion if for all $0 = t_0 < t_1 < \dots < t_m$ the increments $W(t_1) - W(t_0)$, $W(t_2) - W(t_1)$, $\dots$, $W(t_m) - W(t_{m-1})$ are independent and each of these increments is normally distributed with

$$E[W(t_{i+1}) - W(t_i)] = 0 \tag{1.1}$$

$$\text{Var}[W(t_{i+1}) - W(t_i)] = t_{i+1} - t_i \tag{1.2}$$

§1.2.1 Distribution of Brownian motion

Because the increments $W(t_1) - W(t_0)$, $W(t_2) - W(t_1)$, $\dots$, $W(t_m) - W(t_{m-1})$ are independent and normally distributed, the random variables $W(t_1)$, $W(t_2)$, $\dots$, $W(t_m)$ are jointly normally distributed. The joint distribution of jointly normal random variables is determined by their means and covariances. Each of the random

variables $W(t_i)$ has mean zero. For any two times, $0 \leq s < t$, the covariance of $W(s)$ and $W(t)$ is

$$\begin{aligned}\mathbb{E}[W(s)W(t)] &= \mathbb{E}[W(s)(W(t) - W(s)) + W^2(s)] \\ &= \mathbb{E}[W(s)] \cdot \mathbb{E}[W(t) - W(s)] + \mathbb{E}[W^2(s)] \\ &= 0 + \text{Var}[W(s)] = s\end{aligned}$$

Hence, the covariance matrix for Brownian motion (i.e., for the m -dimensional random vector $W(t_1), W(t_2), \dots, W(t_m)$) is

$$\begin{bmatrix} \mathbb{E}[W^2(t_1)] & \mathbb{E}[W(t_1)W(t_2)] & \cdots & \mathbb{E}[W(t_1)W(t_m)] \\ \mathbb{E}[W(t_2)W(t_1)] & \mathbb{E}[W^2(t_2)] & \cdots & \mathbb{E}[W(t_2)W(t_m)] \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}[W(t_m)W(t_1)] & \mathbb{E}[W(t_m)W(t_2)] & \cdots & \mathbb{E}[W^2(t_m)] \end{bmatrix} = \begin{bmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_m \end{bmatrix}$$

The moment-generating function of this random vector $(W(t_1), W(t_2), \dots, W(t_m))$ can be computed using the moment-generating function for a zero-mean normal random variable with variance t and the independence of the increments.

$$\begin{aligned}\varphi(u_1, u_2, \dots, u_m) &= \mathbb{E} \exp \{u_m W(t_m) + u_{m-1} W(t_{m-1}) + \cdots + u_1 W(t_1)\} \\ &= \mathbb{E} \exp \{u_m (W(t_m) - W(t_{m-1}))\} \cdot \mathbb{E} \exp \{(u_{m-1} + u_m) (W(t_{m-1}) - W(t_{m-2}))\} \\ &\quad \cdots \mathbb{E} \exp \{(u_1 + u_2 + \cdots + u_m) W(t_1)\} \\ &= \exp \left\{ \frac{1}{2} u_m^2 (t_m - t_{m-1}) \right\} \cdot \exp \left\{ \frac{1}{2} (u_{m-1} + u_m)^2 (t_{m-1} - t_{m-2}) \right\} \cdots \exp \left\{ \frac{1}{2} (u_1 + u_2 + \cdots + u_m)^2 t_1 \right\}.\end{aligned}$$

§1.2.2 Filtration for Brownian motion

Definition 1.2.2: Filtration for Brownian motion

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space on which is defined a Brownian motion $W(t)$, $t \geq 0$. A filtration for the Brownian motion is a collection of σ -algebras $\mathcal{F}(t)$, $t \geq 0$, satisfying:

- (i) (Information accumulates) For $0 \leq s < t$, every set in $\mathcal{F}(s)$ is also in $\mathcal{F}(t)$. In other words, there is at least as much information available at the later time $\mathcal{F}(t)$ as there is at the earlier time $\mathcal{F}(s)$.
- (ii) (Adaptivity) For each $t \geq 0$, the Brownian motion $W(t)$ at time t is $\mathcal{F}(t)$ -measurable. In other words, the information available at time t is sufficient to evaluate the Brownian motion $W(t)$ at that time.
- (iii) (Independence of future increments) For $0 \leq t < u$, the increment $W(u) - W(t)$ is independent of $\mathcal{F}(t)$. In other words, any increment of the Brownian motion after time t is independent of the information available at time t .

§1.2.3 Martingale Property of Brownian motion

Theorem 1.2.1: Martingale property of Brownian Motion

Brownian Motion is a Martingale

Proof. Let $0 \leq s \leq t$ be given. Then, we have:

$$\mathbb{E}[W(t) \mid \mathcal{F}(s)] = \mathbb{E}[(W(t) - W(s)) + W(s) \mid \mathcal{F}(s)] = \mathbb{E}[W(t) - W(s)] + W(s) = W(s).$$

■

§1.2.4 Quadratic Variation of Brownian motion

Definition 1.2.3: Quadratic Variation of Brownian motion

Let $f(t)$ be a function defined for $0 \leq t \leq T$. The quadratic variation off up to time T is

$$[f, f](T) = \lim_{\|\Pi\| \rightarrow 0} \sum_{j=0}^{n-1} [f(t_{j+1}) - f(t_j)]^2 \quad (1.3)$$

where $\Pi = \{t_0, t_1, \dots, t_n\}$, $\|\Pi\| = \max_{j=0, \dots, n-1} (t_{j+1} - t_j)$. and $0 = t_0 < t_1 < \cdots < t_n = T$.

Theorem 1.2.2

Let W be a Brownian motion. Then

$$[W, W](T) = T$$

for all $T \geq 0$ almost surely.

Proof. Let $\Pi = t_0, t_1, \dots, t_n$ be a partition of $[0, T]$. Define the sampled quadratic variation corresponding to this partition to be

$$Q_\Pi = \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2.$$

We want to show that the quadratic variation has expected value T , and its variance converges to 0. Hence, it converges to its expected value T . We have

$$\mathbb{E}[(W(t_{j+1}) - W(t_j))^2] = \text{Var}[W(t_{j+1}) - W(t_j)] = t_{j+1} - t_j,$$

which implies

$$\mathbb{E}Q_\Pi = \sum_{j=0}^{n-1} \mathbb{E}[(W(t_{j+1}) - W(t_j))^2] = \sum_{j=0}^{n-1} (t_{j+1} - t_j) = T,$$

$$\begin{aligned} \text{Var}[(W(t_{j+1}) - W(t_j))^2] &= \mathbb{E}[(W(t_{j+1}) - W(t_j))^4 - 2(t_{j+1} - t_j)(W(t_{j+1}) - W(t_j))^2 + (t_{j+1} - t_j)^2] \\ &= \mathbb{E}[(W(t_{j+1}) - W(t_j))^4] - 2(t_{j+1} - t_j)\mathbb{E}[(W(t_{j+1}) - W(t_j))^2] + (t_{j+1} - t_j)^2 \\ &= 2(t_{j+1} - t_j)^2, \end{aligned}$$

$$\text{Var}(Q_\Pi) = \sum_{j=0}^{n-1} \text{Var}[(W(t_{j+1}) - W(t_j))^2] = \sum_{j=0}^{n-1} 2(t_{j+1} - t_j)^2 \leq \sum_{j=0}^{n-1} 2\Pi\|(t_{j+1} - t_j) = 2\Pi\|T.$$

Therefore $\lim_{\Pi\| \rightarrow 0} \text{Var}(Q_\Pi) = 0$ and $\lim_{\Pi\| \rightarrow 0} \mathbb{E}(Q_\Pi) = T$. Next, by contradiction, suppose that there exists a set $A \in \mathcal{F}$, with a positive probability $\mathbb{P}(A) > 0$, such that $\forall \omega \in A$ the sample first variation is finite:

$$\sum_{j=0}^{n-1} |W_{t_{j+1}}(\omega) - W_{t_j}(\omega)| < \infty.$$

Then, $\forall \omega \in A$,

$$\sum_{j=0}^{n-1} (W_{t_{j+1}}(\omega) - W_{t_j}(\omega))^2 \leq \underbrace{\max_{0 \leq k \leq n-1} |W_{t_{k+1}}(\omega) - W_{t_k}(\omega)|}_{\rightarrow 0} \cdot \underbrace{\sum_{j=0}^{n-1} |W_{t_{j+1}}(\omega) - W_{t_j}(\omega)|}_{< \infty},$$

which is a contradiction. The first brace follows from the previous result

$$\lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} (W_{t_{j+1}}(\omega) - W_{t_j}(\omega))^2 = T$$

the second brace is due to the assumption that the number n of partition points approaches infinity and the length of the longest subinterval approaches zero, i.e.

$$\lim_{n \rightarrow \infty} \max_{0 \leq k \leq n-1} |W_{t_{k+1}}(\omega) - W_{t_k}(\omega)| = 0,$$

the last brace is because our assumption. ■

To understand better the idea behind Theorem 1.2.2, we choose a large value of n and take $t_j = \frac{jT}{n}$, $j = 0, 1, \dots, n$. Then $t_{j+1} - t_j = \frac{T}{n}$ for all j and

$$(W(t_{j+1}) - W(t_j))^2 = T \cdot \frac{Y_{j+1}^2}{n}.$$

Since the random variables $Y_1, Y_2, \dots, Y_n$ are independent and identically distributed, the Law of Large Numbers implies that $\sum_{j=0}^{n-1} \frac{Y_{j+1}^2}{n}$ converges to the common mean $\mathbb{E}Y_{j+1}^2$ as $n \rightarrow \infty$. This mean is 1, and hence $\sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2$ converges to T . Each of the terms $(W(t_{j+1}) - W(t_j))^2$ in this sum can be quite

different from its mean $t_{j+1} - t_j = \frac{T}{n}$, but when we sum many terms like this, the differences average out to zero. We write informally

$$dW(t) dW(t) = dt \quad (1.4)$$

Let $\Pi = \{t_0, t_1, \dots, t_n\}$ be a partition of $[0, T]$ (i.e., $0 = t_0 < t_1 < \dots < t_n = T$). In addition to computing the quadratic variation of Brownian motion

$$\lim_{\|\Pi\| \rightarrow 0} \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2 = T \quad (1.5)$$

we can compute the cross variation of $W(t)$ with t and the quadratic variation of t with itself, which are

$$\lim_{\|\Pi\| \rightarrow 0} \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j)) (t_{j+1} - t_j) = 0 \quad (1.6)$$

$$\lim_{\|\Pi\| \rightarrow 0} \sum_{j=0}^{n-1} (t_{j+1} - t_j)^2 = 0 \quad (1.7)$$

To see that 0 is the limit in (1.6), we observe that

$$|(W(t_{j+1}) - W(t_j)) (t_{j+1} - t_j)| \leq \max_{0 \leq k \leq n-1} |W(t_{k+1}) - W(t_k)| (t_{j+1} - t_j),$$

and so

$$\left| \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j)) (t_{j+1} - t_j) \right| \leq \max_{0 \leq k \leq n-1} |W(t_{k+1}) - W(t_k)| \cdot T.$$

Since W is continuous, $\max_{0 \leq k \leq n-1} |W(t_{k+1}) - W(t_k)|$ has limit zero as $\|\Pi\|$, the length of the longest subinterval, goes to zero.

$$\lim_{\|\Pi\| \rightarrow 0} \sum_{j=0}^{n-1} (t_{j+1} - t_j)^2 = 0 \quad (1.8)$$

To see that 0 is the limit in (10), we observe that

$$\sum_{j=0}^{n-1} (t_{j+1} - t_j)^2 \leq \max_{0 \leq k \leq n-1} (t_{k+1} - t_k) \cdot \sum_{j=0}^{n-1} (t_{j+1} - t_j) = \|\Pi\| \cdot T,$$

which obviously has limit zero as $\|\Pi\| \rightarrow 0$. Just as we capture (1.5) by writing (1.4), we capture (1.6) and (1.7) by writing

$$dW(t)dt = 0, \quad dtdt = 0.$$

§1.3 Itô Integral

§1.3.1 Itô Integral for simple integrands

To set up the stochastic integral, we will first define the Itô integral for simple integrands and then extend it to general integrands as a limit of the integral of simple integrands.

Definition 1.3.1: simple process

First, we can partition the time interval $[0, T]$ into: $\Pi = \{t_0, t_1, \dots, t_n\}$, where $0 = t_0 \leq t_1 \leq \dots \leq t_n = T$. A **simple process** $\Delta(t)$ is defined as a stochastic process that takes constant values on each subinterval $[t_j, t_{j+1})$, having the following properties:

- The value of the simple process at a time point t_j can only depend on information available up to that time point
- The value of $\Delta(t)$ on the second interval $[t_1, t_2)$ can depend on observations made during the first time interval $[0, t_1)$

Starting with Itô's Integral for Simple Integrands, our goal is to make sense of the equation below.

$$\int_0^T \Delta(t) dW(t). \quad (1.9)$$

The basic ingredients here are a Brownian motion $W(t)$, $t \geq 0$, together with a filtration $\mathcal{F}(t)$, $t \geq 0$, for this Brownian motion. We will let the integrand $\Delta(t)$ be a simple process.

Construction: We shall interpret the interaction between the simple process $\Delta(t)$ and the Brownian motion $W(t)$ through a financial framework:

- Let $W(t)$ represent the price per share of an asset at time t (ignoring that Brownian motion can be negative)
- Consider $t_0, t_1, \dots, t_{n-1}$ as discrete trading dates
- Define $\Delta(t_j)$ as the position (number of shares) held from time t_j until t_{j+1}

This structure provides an intuitive foundation for constructing the Itô integral. The cumulative trading gain $I(t)$ at time t is defined recursively:

$$\begin{aligned} \text{For } 0 \leq t \leq t_1 : \quad I(t) &= \Delta(0)[W(t) - W(0)] \\ \text{For } t_1 \leq t \leq t_2 : \quad I(t) &= I(t_1) + \Delta(t_1)[W(t) - W(t_1)] \end{aligned}$$

Generalizing, for any t where $t_k \leq t \leq t_{k+1}$:

$$I(t) = \sum_{j=0}^{k-1} \Delta(t_j)[W(t_{j+1}) - W(t_j)] + \Delta(t_k)[W(t) - W(t_k)]$$

This process $I(t)$ defines the Itô integral for simple processes:

$$I(t) = \int_0^t \Delta(u) dW(u)$$

Setting $t = T$ provides the complete integral over $[0, T]$, while intermediate values of t give the integral over $[0, t]$, offering a full definition of the stochastic integral process.

Properties: Linearity, Martingale Property, Itô Isometry, Quadratic Variation (accumulated at rate $\Delta^2(t)$ per unit time). Here we will prove the last two properties.

Theorem 1.3.1: Itô Isometry

The Itô integral defined by (1.9) satisfies

$$\mathbb{E}I^2(t) = \mathbb{E} \int_0^t \Delta^2(u) du.$$

Proof. To simplify the notation, we set $D_j = W(t_{j+1}) - W(t_j)$ for $j = 0, \dots, k-1$ and $D_k = W(t) - W(t_k)$ so that (1.9) may be written as $I(t) = \sum_{j=0}^k \Delta(t_j)D_j$ and

$$I^2(t) = \sum_{j=0}^k \Delta^2(t_j)D_j^2 + 2 \sum_{0 \leq i < j \leq k} \Delta(t_i)\Delta(t_j)D_iD_j.$$

We first show that the expected value of each of the cross terms is zero. For $i < j$, the random variable $\Delta(t_i)\Delta(t_j)D_i$ is $\mathcal{F}(t_j)$ -measurable, while the Brownian increment D_j is independent of $\mathcal{F}(t_j)$. Furthermore, $\mathbb{E}D_j = 0$. Therefore,

$$\mathbb{E}[\Delta(t_i)\Delta(t_j)D_iD_j] = \mathbb{E}[\Delta(t_i)\Delta(t_j)D_i] \cdot \mathbb{E}D_j = \mathbb{E}[\Delta(t_i)\Delta(t_j)D_i] \cdot 0 = 0.$$

We next consider the square terms $\Delta^2(t_j)D_j^2$. The random variable $\Delta^2(t_j)$ is $\mathcal{F}(t_j)$ -measurable, and the squared Brownian increment D_j^2 is independent of $\mathcal{F}(t_j)$. Furthermore, $\mathbb{E}D_j^2 = t_{j+1} - t_j$ for $j = 0, \dots, k-1$ and $\mathbb{E}D_k^2 = t - t_k$. Therefore,

$$\begin{aligned} \mathbb{E}I^2(t) &= \sum_{j=0}^k \mathbb{E}[\Delta^2(t_j)D_j^2] = \sum_{j=1}^k \mathbb{E}\Delta^2(t_j) \cdot \mathbb{E}D_j^2 \\ &= \sum_{j=0}^{k-1} \mathbb{E}\Delta^2(t_j)(t_{j+1} - t_j) + \mathbb{E}\Delta^2(t_k)(t - t_k). \end{aligned}$$

But $\Delta(t_j)$ is constant on the interval $[t_j, t_{j+1})$, and hence $\Delta^2(t_j)(t_{j+1} - t_j) = \int_{t_j}^{t_{j+1}} \Delta^2(u) du$. Similarly, $\Delta^2(t_k)(t - t_k) = \int_{t_k}^t \Delta^2(u) du$.

$t_k) = \int_{t_k}^t \Delta^2(u) du$. We may thus obtain

$$\begin{aligned} \mathbb{E}I^2(t) &= \sum_{j=0}^{k-1} \mathbb{E} \int_{t_j}^{t_{j+1}} \Delta^2(u) du + \mathbb{E} \int_{t_k}^t \Delta^2(u) du \\ &= \mathbb{E} \left[\sum_{j=0}^{k-1} \int_{t_j}^{t_{j+1}} \Delta^2(u) du + \int_{t_k}^t \Delta^2(u) du \right] = \mathbb{E} \int_0^t \Delta^2(u) du. \end{aligned}$$

■

Theorem 1.3.2: Quadratic Variation of the Itô Integral

Let

$$I(t) = \int_0^t \Delta(u) dW(u)$$

be the Itô integral with a simple integrand $\Delta(\cdot)$. Then the quadratic variation of $I(t)$ is

$$[I, I](t) = \int_0^t \Delta^2(u) du.$$

Proof. Since $\Delta(u)$ is simple, there exists a partition

$$0 = t_0 < t_1 < \cdots < t_N = t,$$

such that $\Delta(u)$ is constant on every interval $[t_j, t_{j+1})$. Denote the constant value on $[t_j, t_{j+1})$ by $\Delta(t_j)$. Then, on each such interval, the Itô integral can be written as

$$I(t) - I(t_j) = \Delta(t_j) \left(W(t) - W(t_j) \right) \quad \text{for } t \in [t_j, t_{j+1}).$$

To compute the quadratic variation over one subinterval, fix an interval $[t_j, t_{j+1}]$ and consider a finer partition

$$t_j = s_0 < s_1 < \cdots < s_m = t_{j+1}.$$

On this subinterval, the increment of the Itô integral is

$$I(s_{i+1}) - I(s_i) = \Delta(t_j) \left(W(s_{i+1}) - W(s_i) \right), \quad i = 0, \dots, m-1.$$

Thus, the sum of the squared increments becomes

$$\begin{aligned} \sum_{i=0}^{m-1} \left[I(s_{i+1}) - I(s_i) \right]^2 &= \sum_{i=0}^{m-1} \left[\Delta(t_j) \left(W(s_{i+1}) - W(s_i) \right) \right]^2 \\ &= \Delta^2(t_j) \sum_{i=0}^{m-1} \left(W(s_{i+1}) - W(s_i) \right)^2. \end{aligned} \tag{1.10}$$

Since Brownian motion has quadratic variation equal to the elapsed time, as the mesh of the partition tends to zero (i.e. $m \rightarrow \infty$ and $\max_i (s_{i+1} - s_i) \rightarrow 0$), the sum in (1.10) converges to the length of the interval:

$$\sum_{i=0}^{m-1} \left(W(s_{i+1}) - W(s_i) \right)^2 \longrightarrow t_{j+1} - t_j.$$

Hence, the quadratic variation accumulated on the subinterval $[t_j, t_{j+1}]$ is

$$\Delta^2(t_j)(t_{j+1} - t_j) = \int_{t_j}^{t_{j+1}} \Delta^2(u) du,$$

where the equality follows from the constancy of $\Delta(u)$ on this interval.

By summing over all subintervals in the partition of $[0, t]$, we obtain

$$[I, I](t) = \sum_{j=0}^{N-1} \Delta^2(t_j)(t_{j+1} - t_j) = \int_0^t \Delta^2(u) du.$$

This completes the proof of Theorem 1.3.2. ■

§1.3.2 Itô Integral for general integrands

Definition 1.3.2: general integrands

For general integrands $\Delta(t)$, we no longer require them to be simple processes (piecewise constant), but instead allow:

- Continuous variation over time
- Jumps in the process
- Subject to two key conditions:
 1. Adaptedness: $\Delta(t)$ is adapted to the filtration $\mathcal{F}(t)$
 2. Square-integrability: $\mathbb{E} \int_0^T \Delta^2(t) dt < \infty$

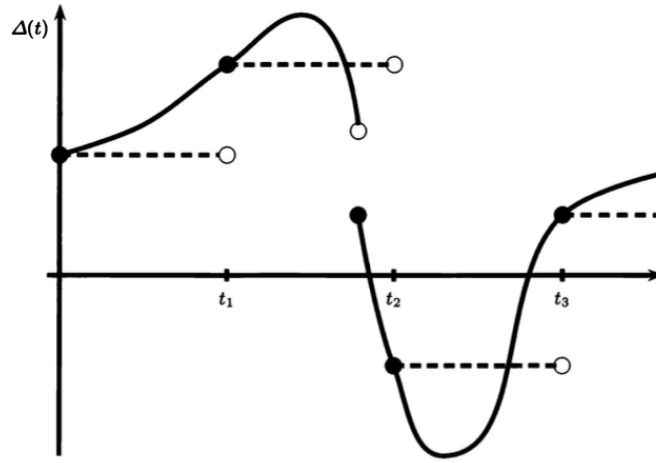


Figure 1.1: Approximation of a general process by a simple process

As illustrated in Figure 1.1, the idea is to finer the partition (maximum step size approaching zero) to let the simple process increasingly better approximate the original process

Let $\Delta(t)$ be a progressively measurable integrand defined on $[0, T]$ satisfying the square-integrability. It is known that there exists a sequence $\{\Delta_n(t)\}_{n \geq 1}$ of simple processes satisfying

$$\lim_{n \rightarrow \infty} \mathbb{E} \int_0^T |\Delta_n(t) - \Delta(t)|^2 dt = 0. \quad (1.11)$$

This convergence is in the L^2 -sense on $[0, T] \times \Omega$, meaning that $\Delta_n(t)$ converges to $\Delta(t)$ in mean square. For each simple process $\Delta_n(t)$, the Itô integral is defined by

$$I_n(t) = \int_0^t \Delta_n(u) dW(u), \quad 0 \leq t \leq T. \quad (1.12)$$

By Itô isometry (Theorem (1.3.1)), which for any two simple processes $\Delta_n(t)$ and $\Delta_m(t)$ gives:

$$\mathbb{E} [I_n(t) - I_m(t)]^2 = \mathbb{E} \int_0^t |\Delta_n(u) - \Delta_m(u)|^2 du. \quad (1.13)$$

We now define the Itô integral for the general integrand $\Delta(t)$ by taking the limit of the integrals corresponding to the approximating sequence:

$$\int_0^t \Delta(u) dW(u) = \lim_{n \rightarrow \infty} \int_0^t \Delta_n(u) dW(u), \quad 0 \leq t \leq T. \quad (1.14)$$

Our goal is to show that the limit on the right-hand side exists in the space $L^2(\Omega)$.

Proof of the Limit Existence: To prove the existence of the limit, consider any two indices $n, m \geq 1$. By the

Itô isometry (1.13), we have:

$$\mathbb{E} [|I_n(t) - I_m(t)|^2] = \mathbb{E} \int_0^t |\Delta_n(u) - \Delta_m(u)|^2 du.$$

Since $\Delta_n(t)$ converges to $\Delta(t)$ in L^2 , for every $\varepsilon > 0$ there exists an $N \in \mathbb{N}$ so that for all $n, m \geq N$,

$$\mathbb{E} \int_0^T |\Delta_n(u) - \Delta_m(u)|^2 du < \varepsilon.$$

Thus, it follows that

$$\mathbb{E} [|I_n(t) - I_m(t)|^2] < \varepsilon, \quad \text{for all } 0 \leq t \leq T.$$

This demonstrates that the sequence $\{I_n(t)\}$ is Cauchy in $L^2(\Omega)$. Since the space $L^2(\Omega)$ is complete, every Cauchy sequence converges to a limit in $L^2(\Omega)$. Therefore, there exists a process $I(t) \in L^2(\Omega)$ such that

$$\lim_{n \rightarrow \infty} \mathbb{E} [|I_n(t) - I(t)|^2] = 0.$$

This limit is independent of the particular approximating sequence (provided the approximations satisfy (1.11)). We then define

$$I(t) = \int_0^t \Delta(u) dW(u),$$

which completes the definition of the Itô integral for continuously varying integrands.

Remark 1: Inheritance of Properties

Since any continuously varying integrand $\Delta(t)$ satisfying square-integrability can be approximated in L^2 by a sequence of simple processes $\{\Delta_n(t)\}$, the corresponding Itô integrals

$$I_n(t) = \int_0^t \Delta_n(u) dW(u)$$

inherit key properties such as Martingale Property, Itô Isometry (Theorem (1.3.1)) and Quadratic Variation (Theorem (1.3.2)). Because the L^2 -limit

$$I(t) = \lim_{n \rightarrow \infty} I_n(t)$$

exists in the complete space $L^2(\Omega)$ and the L^2 -norm is continuous with respect to these operations, these properties carry over to the limit. In other words, the general Itô integral retains the structure and characteristic properties of the integrals defined for simple processes.

§1.4 Itô-Doeblin Formula

For a differentiable function $f(x)$ applied to Brownian motion $W(t)$, the classical chain rule fails because Brownian motion is not differentiable. Instead, the Itô-Doeblin formula provides the correct differentiation rule:

Theorem 1.4.1: Itô-Doeblin formula for Brownian motion

Let $f(t, x)$ be a function with continuous partial derivatives $f_t(t, x)$, $f_x(t, x)$, and $f_{xx}(t, x)$, and let $W(t)$ be a Brownian motion. Then, for every $T \geq 0$,

$$f(T, W(T)) = f(0, W(0)) + \int_0^T f_t(t, W(t)) dt + \int_0^T f_x(t, W(t)) dW(t) + \frac{1}{2} \int_0^T f_{xx}(t, W(t)) dt$$

Proof. Let $\Pi = \{t_0, t_1, \dots, t_n\}$ be a partition of the interval $[0, T]$. By Taylor's theorem, for each j we have

$$\begin{aligned} f(t_{j+1}, S(t_{j+1})) - f(t_j, S(t_j)) &= f_t(t_j, S(t_j)) (t_{j+1} - t_j) + f_x(t_j, S(t_j)) (S(t_{j+1}) - S(t_j)) \\ &\quad + \frac{1}{2} f_{xx}(t_j, S(t_j)) (S(t_{j+1}) - S(t_j))^2 + \frac{1}{2} f_{tt}(t_j, S(t_j)) (t_{j+1} - t_j)^2 \\ &\quad + f_{tx}(t_j, S(t_j)) (t_{j+1} - t_j) (S(t_{j+1}) - S(t_j)) + \text{h.o.t.} \end{aligned}$$

Summing over $j = 0, 1, \dots, n-1$, we obtain

$$\begin{aligned} f(T, S(T)) - f(0, S(0)) &= \sum_{j=0}^{n-1} f_t(t_j, S(t_j))(t_{j+1} - t_j) + \sum_{j=0}^{n-1} f_x(t_j, S(t_j))(S(t_{j+1}) - S(t_j)) \\ &\quad + \frac{1}{2} \sum_{j=0}^{n-1} f_{xx}(t_j, S(t_j))(S(t_{j+1}) - S(t_j))^2 + \frac{1}{2} \sum_{j=0}^{n-1} f_{tt}(t_j, S(t_j))(t_{j+1} - t_j)^2 \\ &\quad + \sum_{j=0}^{n-1} f_{tx}(t_j, S(t_j))(t_{j+1} - t_j)(S(t_{j+1}) - S(t_j)) + \text{h.o.t.} \end{aligned}$$

Taking the limit as $\|\Pi\| \rightarrow 0$, we have:

- **First Term:** it is a Riemann sum converging to the Lebesgue integral:

$$\sum_{j=0}^{n-1} f_t(t_j, S(t_j))(t_{j+1} - t_j) \rightarrow \int_0^T f_t(t, S(t)) dt.$$

- **Second Term:** it converges, by definition, to the Itô integral:

$$\sum_{j=0}^{n-1} f_x(t_j, S(t_j))(S(t_{j+1}) - S(t_j)) \rightarrow \int_0^T f_x(t, S(t)) dS(t).$$

- **Third Term:** By the property of quadratic variation,

$$\frac{1}{2} \sum_{j=0}^{n-1} f_{xx}(t_j, S(t_j))(S(t_{j+1}) - S(t_j))^2 \rightarrow \frac{1}{2} \int_0^T f_{xx}(t, S(t)) d\langle S, S \rangle(t).$$

- **Fourth Term and higher order terms:** Fourth term is of the form

$$\frac{1}{2} \sum_{j=0}^{n-1} f_{tt}(t_j, S(t_j))(t_{j+1} - t_j)^2$$

, which consists of terms of order $(\Delta t)^2$. Since $\sum_{j=0}^{n-1} (t_{j+1} - t_j) = T$, the entire sum is bounded by a constant times $\|\Pi\|$ and vanishes as $\|\Pi\| \rightarrow 0$. Higher order terms vanish for the same reason.

Thus, Itô's formula is given by

$$f(T, S(T)) - f(0, S(0)) = \int_0^T f_t(t, S(t)) dt + \int_0^T f_x(t, S(t)) dS(t) + \frac{1}{2} \int_0^T f_{xx}(t, S(t)) d\langle S, S \rangle(t).$$

■

Definition 1.4.1: Itô Process

Itô processes are stochastic processes more general than Brownian motion (eg. geometric Brownian motion). Let $W(t)$, $t \geq 0$, be a Brownian motion, and let $\mathcal{F}(t)$, $t \geq 0$, be an associated filtration. An Itô process is a stochastic process of the form:

$$X(t) = X(0) + \int_0^t \Delta(u) dW(u) + \int_0^t \Theta(u) du$$

where $X(0)$ is nonrandom, and $\Delta(u)$ and $\Theta(u)$ are adapted stochastic processes.

Theorem 1.4.2: Itô-Doeblin Formula for Itô Process

Let $X(t)$, $t \geq 0$, be an Itô process, and let $f(t, x)$ be a function for which the partial derivatives $f_t(t, x)$, $f_x(t, x)$, and $f_{xx}(t, x)$ are defined and continuous. Then, for every $T \geq 0$,

$$f(T, X(T)) = f(0, X(0)) + \int_0^T f_t(t, X(t)) dt + \int_0^T f_x(t, X(t)) dX(t) + \frac{1}{2} \int_0^T f_{xx}(t, X(t)) d[X, X](t)$$

Recast the Itô Formula in differential notation, and we can get:

$$df(t, X(t)) = f_t(t, X(t)) dt + f_x(t, X(t))\Delta(t) dW(t) + f_x(t, X(t))\Theta(t) dt + \frac{1}{2}f_{xx}(t, X(t))\Delta^2(t) dt$$

Comments: The proof of a more general version of Itô-Doeblin formula is similar to the proof of Theorem (1.4.1) despite the Brownian motion $S(t)$ is replaced by an Itô process $X(t)$. One also need to prove first the quadratic variation of X is $\int_0^t \sigma^2(u) du$ by decomposing the Itô process into its continuous drift and stochastic integral parts and showing that the drift and cross-variation terms vanish as the partition mesh tends to zero

§1.5 Black-Scholes Equation

Consider a market having a money market account and a stock. The money account pays a constant rate of interest r and the stock is modeled by a geometric Brownian motion

$$dS_t = \alpha S_t dt + \sigma S_t dW_t.$$

Suppose that there is a trader who at each time t has a portfolio valued at X_t . He buys Δ_t shares of the stock and invests $X_t - \Delta_t S_t$ in the money market account. Then, X_t follows

$$dX_t = rX_t dt + \Delta_t(\alpha - r)S_t dt + \Delta_t \sigma S_t dW_t.$$

Consider a European call option that pays $(S_T - K)^+$ at time T . We want to calculate its price at time $t < T$. Denote it by $c(t, x)$, which is the price of the option at time t if the stock price then is $S_t = x$. By Itô-Doeblin Formula,

$$d(e^{-rt}c(t, S_t)) = e^{-rt} \left[-rc(t, S_t) + c_t(t, S_t) + \alpha S_t c_x(t, S_t) + \frac{1}{2}\sigma^2 S_t^2 c_{xx}(t, S_t) \right] dt + e^{-rt} \sigma S_t c_x(t, S_t) dW_t$$

On the other hand, a short option hedging portfolio starts with some initial capital X_0 and invest in the stock and money market account so that the portfolio value X_t at each time matches with $c(t, S_t)$. In other words, $X_0 = c(0, S_0)$ and

$$d(e^{-rt}c(t, S_t)) = d(e^{-rt}X_t).$$

Applying Ito formula on the right hand side, we have

$$d(e^{-rt}X_t) = (\alpha - r)\Delta_t e^{-rt}S_t dt + \sigma \Delta_t e^{-rt}S_t dW_t.$$

This leads to the Black-Scholes-Merton partial differential equation

$$rc(t, x) = c_t(t, x) + rxc_x(t, x) + \frac{1}{2}\sigma^2 x^2 c_{xx}(t, x)$$

and $c(T, x) = (x - K)^+$.

Theorem 1.5.1: Black-Scholes-Merton Formula

The solution to the above PDE is given by

$$c(t, x) = xN(d_+(T-t, x)) - Ke^{-r(T-t)}N(d_-(T-t, x))$$

where

$$d_{\pm} = \frac{1}{\sigma\sqrt{T-t}} \left[\log \frac{x}{K} + \left(r \pm \frac{\sigma^2}{2} \right) (T-t) \right]$$

and N is the cumulative standard normal distribution function.